# Neuroscience: The Predictive Brain and the Body That Feels Lack

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# Neuroscience: The Predictive Brain and the Body That Feels Lack

## 1. Purpose of the Investigation

Figure 1: Neuroscience: The Predictive Brain and the Body That Feels Lack —

This report presents the results of an investigation into whether neuroscience shows structural correspondence with the five-stage model of creation, Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ), and with four associated definitions concerning prediction error, subjective lack, the retention of unresolved discrepancy, and the construction of emotion.

Neuroscience occupies a core position within the thirty domains investigated here. Every one of the associated definitions, from the stance of taking up prediction error as a question, to the subjective experience of discrepancy as lack, to the function of retaining discrepancy without reflexively processing it, to the way discrepancy is evaluated and constructed as emotion, finds a direct frame of reference in neuroscience. This document is an investigation report. It does not present neuroscientific facts as the "grounds" of the theory, but as corresponding structures.

## 2. Method of Investigation

In the neuroscience domain, eleven theories and phenomena were selected for evaluation. The selection criterion was that they be established theories, or actively debated frameworks, concerning prediction, error, learning, consciousness, emotion, and embodiment. The set includes computational levels (predictive processing, the free-energy principle), physiological levels (synaptic plasticity, rhythmic synchrony, neural avalanches), bodily and emotional levels (interoception, constructionist theories of emotion, polyvagal theory, embodied cognition), levels of cognitive control (executive function, theories of consciousness), and research on sleep and dreams.

Evaluation consisted in examining the correspondence of each theory with the five-stage model

and with the four associated definitions, while giving particular weight to the fit with the three conditions of Edge: relational network, indeterminacy, and connection to Vortex.

## 3. Overview of the Five-Stage Model

The five-stage model referenced in this investigation is explained below. It is a model for describing the generation of being itself and is not restricted to any particular scale or domain.

**Field** - An undifferentiated state. It is the initial condition in which neither direction nor structure has yet been determined, a zero-state in which nothing has yet occurred.

**Wave** - A stage of exploration in which multiple directions diverge and compete. Differences arise within what had previously been undifferentiated, and opposition and fluctuation appear.

**Edge** - A state of tension in which opposing elements coexist without converging on either side. This is the core stage of the model, and the retention of tension at the boundary is taken to be the essence of creation.

**Vortex** - A stage in which a new coherence or order spontaneously emerges from within tension. The contradiction retained at Edge is reorganized into a higher-order structure. In neuroscience, this appears as the updating of predictive models, self-reinforcing loops in synaptic connectivity, phase-locking entrainment, and the construction of emotional categories.

**Bundle** - A stage in which form becomes determinate and stabilizes as a reusable structure. In neuroscience, this is realized as updated predictive models, new synaptic connectivity patterns, consciously formed emotional experience, and stable self-models. Bundle is not fixed; it becomes the precondition of a new Field.

If anything in these definitions feels uneasy, that itself is an important perspective for this investigation. Continuing to read while retaining that unease, or leaving the report at this point, are both meaningful choices.

## 4. Findings

### Overall Picture

Of the eleven neuroscientific theories evaluated, ten showed structural correspondence. One, neural avalanches and criticality, was judged to be conditionally corresponding. As a core domain, neuroscience contains direct frames of reference for all four associated definitions.

The greatest strength of the neuroscience domain is the density of correspondence across three layers. The computational layer (predictive processing), the physiological layer (synaptic plasticity, rhythmic synchrony), and the bodily-emotional layer (interoception, emotion construction, the polyvagal framework) each have independent correspondences with the five stages, and these layers are also interconnected.

### Major Findings

**Predictive Processing: A Computational Frame for Error-Driven Thought** According to predictive processing, the brain is a machine that minimizes prediction error through hierarchical Bayesian inference. Higher layers send predictions downward, and lower layers return error upward; this bidirectional processing offers a unified description of perception, action, and learning. Precision controls the weighting of error and thereby determines which errors are taken up.

This framework functions as the computational frame of reference for error-driven thought. The idea of taking up an error that would ordinarily be discarded as a question corresponds to the

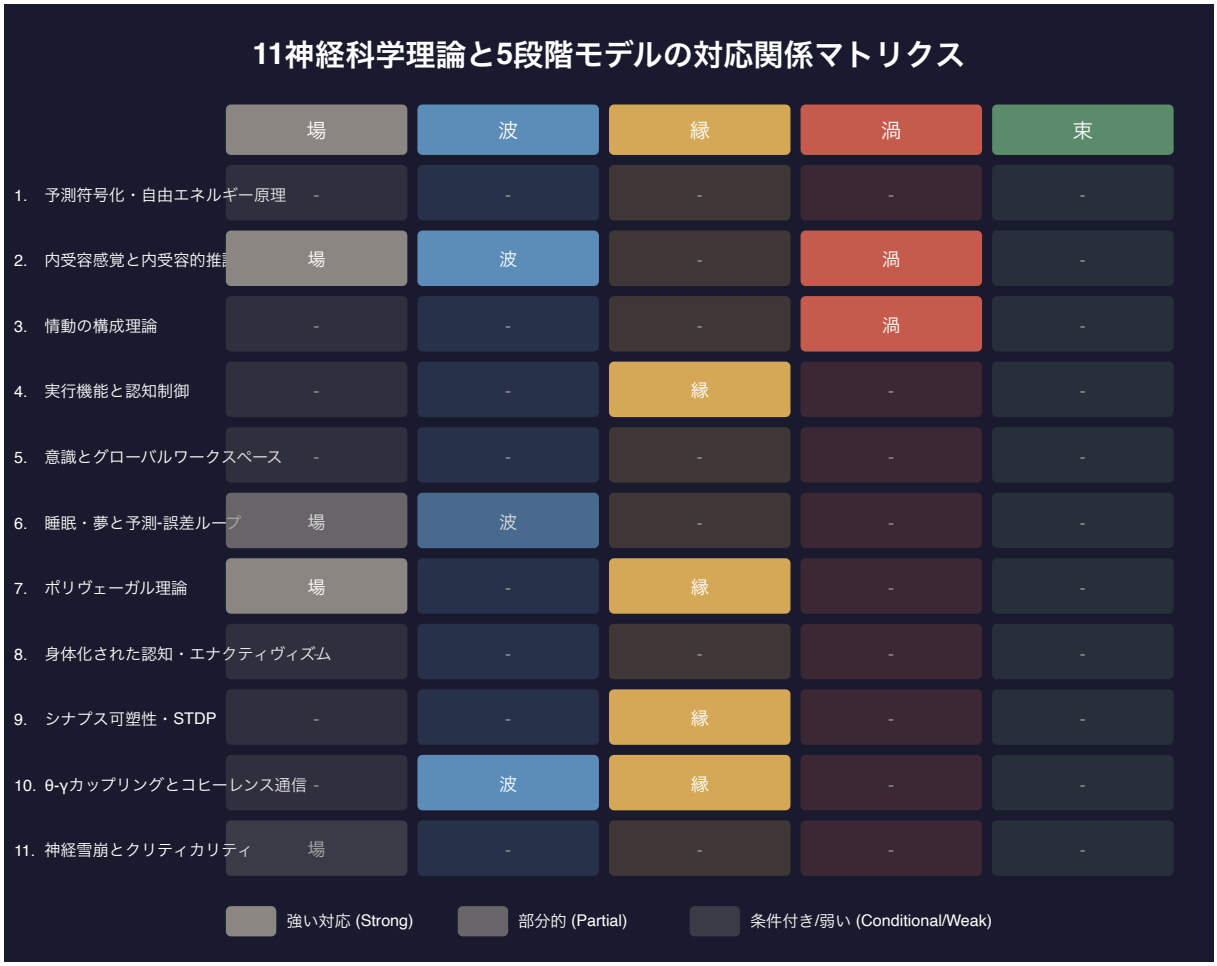


Figure 2: Neuroscience: The Predictive Brain and the Body That Feels Lack — ×5

fact that low-precision prediction errors are normally discarded. The retention of unresolved discrepancy may function here as an intentional manipulation of precision.

At the same time, this framework does not exhaust the whole theory. The first-person dimension of discrepancy as something "felt as lacking" cannot be captured by computation alone, and the retention of unresolved discrepancy includes semantic holding, not merely a change in precision weighting.

**Synaptic Plasticity: Timing Decides Fate** STDP (spike-timing-dependent plasticity) is the mechanism by which synaptic efficacy is strengthened or weakened according to the relative timing of presynaptic input and postsynaptic spikes. A relational quantity, a time difference of tens of milliseconds, directly determines the direction of structural change.

At this level, one can clearly observe, without consciousness, the process by which a physical Wave, a train of spikes, is mapped through Edge, a temporal relation, into Bundle, a connectivity pattern. The essence of Edge, namely that relation determines direction, becomes directly observable at the synaptic level.

**Rhythmic Synchrony: Coupling Becomes the Condition of Communication** Neural communication through theta-gamma coupling and coherence, as described by the CTC framework, characterizes the mechanism by which neural populations in distinct brain regions become connected through phase synchrony and thereby make information transfer possible. Independent of the temporal lag emphasized by STDP, a frequency-based relational quantity, phase synchrony, becomes the condition for communication.

The fact that these two lineages, STDP and CTC, both implement the same structure, relation generates structure, through different physical quantities, suggests that Edge is not a particular mechanism but a structural feature.

**Interoception and the Construction of Emotion: The Body Feels Lack** Predictive processing has also been applied to interoception, the perception of heartbeat, breathing, the gut, and other internal bodily processes, and interoceptive prediction error is experienced as bodily discrepancy. This interoceptive prediction error is the bodily basis for the subjective experience in which discrepancy between prediction and reality is taken up as lack.

The constructionist theory of emotion (Barrett 2017) describes emotion as an active process of construction by the brain. The integration and evaluation of interoceptive prediction with exteroceptive context provides a frame of reference for the evaluative dimension, and the construction of emotional categories corresponds to Vortex, the emergence of a new coherence.

**Polyvagal Theory: Nothing Can Be Retained Without Safety** Polyvagal theory describes a three-stage structure of the autonomic nervous system, dorsal vagal, sympathetic, and ventral vagal. Unless the ventral vagal system is active, that is, unless a safe state is present, the retention of unresolved discrepancy cannot function. Under threat, sympathetic dominance, fight or flight, or dorsal vagal dominance, freezing or shutdown, takes over, and the margin needed to hold a question without reflexive discharge is lost.

Co-regulation through social engagement is itself a structure in which one contacts and is affected by another at the boundary, a directly interpersonal realization of Edge. The neuroanatomical details of the theory remain debated, and that contested status is reported here as such.

**Embodied Cognition: Lack Is a Bodily Process** 4E cognition (Embodied, Embedded, Enacted, Extended) and enactivism argue that cognition arises not from information processing

inside the brain alone but from a coupled body-environment system. The membrane of autopoiesis, the boundary between self and non-self, is an especially close counterpart to a core description of Edge.

This framework provides the philosophical ground for the claim that lack is first something felt by the body before it becomes something thought through conceptually.

## 5. Correspondence with the Five Stages

### Field: Predictive Models and the Conditions of the Field

Existing hierarchical generative models in predictive processing, bodily homeostasis, safe relational fields enabled by ventral-vagal activation, and circuit states near criticality all correspond to Field. In neuroscience, Field is multilayered and functions as a computational, physiological, and social precondition.

### Wave: Prediction Error and Fluctuation

Mismatch between prediction and sensory input, discrepancy in interoceptive prediction, spike trains as temporal fluctuations, and rhythmic oscillations in the brain such as theta and gamma all correspond to Wave. Wave appears in diverse forms at every scale of neuroscience.

### Edge: Selection Through Relational Coupling

The weighting of error through precision, the relative timing of STDP, the phase synchrony of CTC, multimodal integration in the insula, and the contact point of social engagement all correspond to Edge. The multiple implementation of Edge is a striking feature of neuroscience. The time lag of STDP, as a temporal relation, the phase synchrony of CTC, as a frequency relation, and social engagement in polyvagal theory, as an interpersonal relation, each independently implement the same structure: relation determines direction. Four cases showed especially strong correspondence.

### Vortex: Model Updating and Construction

Reorganization of predictive models, self-reinforcing loops of plasticity, phase-locking entrainment, construction of emotional categories, and stabilization through co-regulation all correspond to Vortex. Vortex operates, with or without consciousness, as the dynamic by which error reorganizes an existing structure.

### Bundle: Stable New Structure

Updated predictive models, as the seed of the next Field, new synaptic connectivity patterns, as traces of memory, coordinated information patterns, formed emotional experience, and the establishment of a secure base all correspond to Bundle. In neuroscience, Bundle has a cyclical character, functioning as the Field of the next five-stage cycle.

## 6. Insights From This Domain

### Creation Is Not the Privilege of Consciousness

Synaptic plasticity performs the sequence time lag -> structural change without consciousness, rhythmic synchrony performs synchrony -> selective communication without consciousness, and in dreams predictive and error loops continue to operate even without higher-order cognition. The five stages operate isomorphically in both conscious and non-conscious cases.

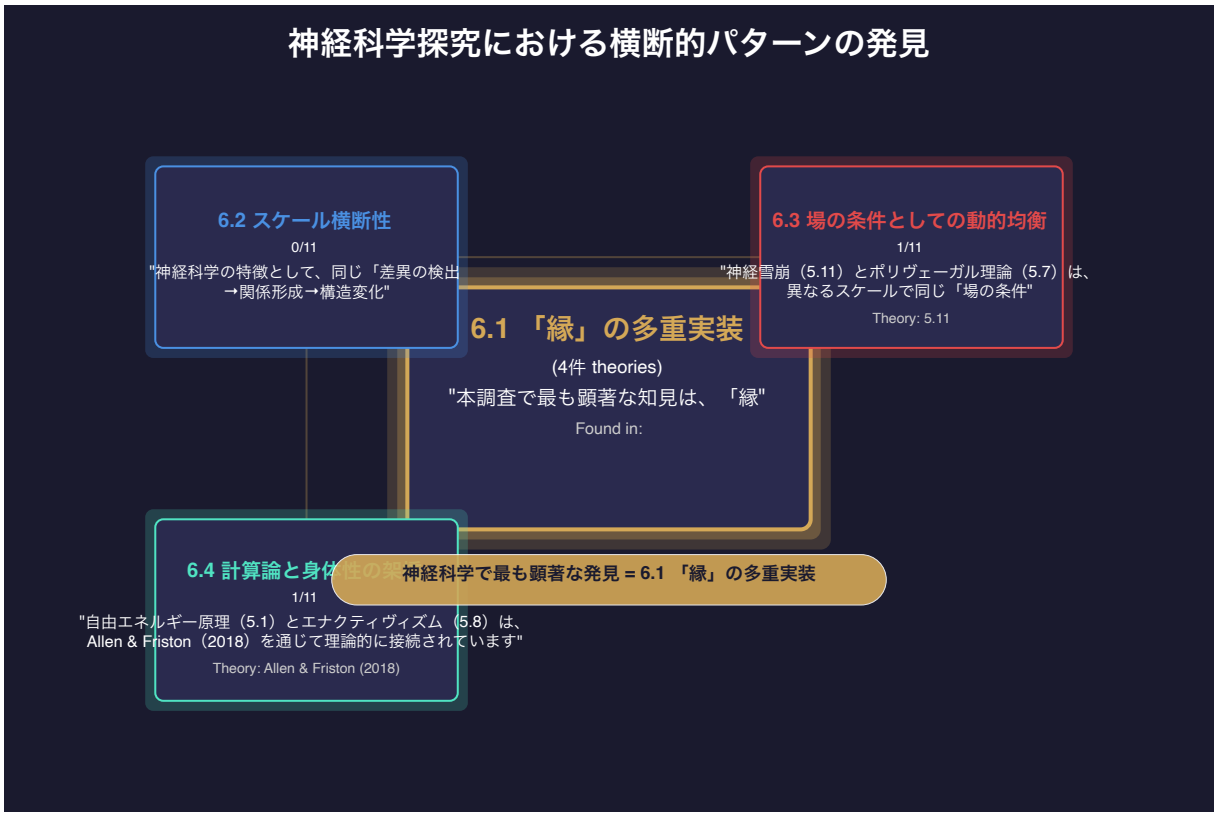


Figure 3: Neuroscience: The Predictive Brain and the Body That Feels Lack —

This suggests that error-driven thought is, before being a conscious method of thinking, a basic mode of operation of the nervous system. At the same time, the definition centered on subjective experience remains limited to conscious levels, and a distinction is needed: non-conscious processes belong within the descriptive scope of the five stages, but not within the narrower scope of that subjective definition.

### Multiple Implementations of Edge

STDP (time lag), CTC (phase synchrony), polyvagal theory (social engagement), and the insula (multimodal integration) together make neuroscience a treasure house of multiple implementations of Edge. All share the structure that relation determines direction, while differing completely in physical realization. Edge is not a specific mechanism; it is a structural feature of selection through relation.

### A Sense of Safety Is a Precondition for Creation

The polyvagal framework shows that, without a safe Field, the retention of unresolved discrepancy cannot function. Under threat, people either discharge immediately into processing, fight or flight, or freeze and shut down. The finding that one cannot create without safety has direct implications for education, clinical practice, and organizational design.

### Neuroscientific Support for the PID Correspondence

The correspondence found in D07 between PID control and the associated error-driven structure is supported from the side of neuroscience as well. Precision control in predictive processing, prediction error itself, and detection of the rate of change of prediction error define a structure that is independently confirmed in both engineering and neuroscience.

## 7. Unresolved Questions

- **The range of consciousness**: If the five stages operate even at levels that do not involve consciousness, how far can the phrase "taken up by consciousness" be extended? Is there anything like an experience of lack at the level of synaptic plasticity? This question simultaneously approaches and resists panpsychist interpretations.
- **An integrated model of retained discrepancy**: Executive function, precision control, and safety regulation each describe different aspects of retaining unresolved discrepancy. It remains unclear whether these are aspects of a single integrated mechanism or the outcome of multiple independent processes.
- **Formalizing Edge**: How should the structural feature of selection through relation be formalized? A language is needed that can encompass the diverse implementations of time lag, phase synchrony, and social engagement.
- **Critiques of polyvagal theory**: The broad framework is often regarded as clinically useful, but the degree of functional differentiation within the vagal system and the strictness of its evolutionary claims remain debated.

## 8. Summary

Neuroscience is a core domain in this investigation, with a high rate of correspondence: ten of the eleven cases showed substantial alignment. It contains direct frames of reference for all four associated definitions, and it is a domain in which the same structure repeats fractally from synapses to interpersonal relations.

The highest-confidence finding is that predictive processing provides the computational frame of reference for all four associated definitions. The cycle prediction error generation -> precision weighting -> model updating -> new prediction is a determinate structural correspondence spanning all stages of the five-stage model.

Strong findings also include the proposition that creation is not the privilege of consciousness, and the multiple implementation of Edge. From the synaptic level to the interpersonal level, the structure that relation determines direction recurs across several independent lineages.

More tentatively, the claim that a sense of safety is a precondition for retaining unresolved discrepancy remains a hypothesis with direct practical implications for education and clinical work. The details of the polyvagal framework remain debated, but the broader insight is practically significant.

Unresolved questions remain concerning the range of consciousness and the integrated model of retained discrepancy. These touch the core of the theory and are not questions that should be solved in haste.